



1.12

Transformations of Functions

Student Notes and Guided Practice

AP Precalculus - Unit 1

| ECHS Mathematics

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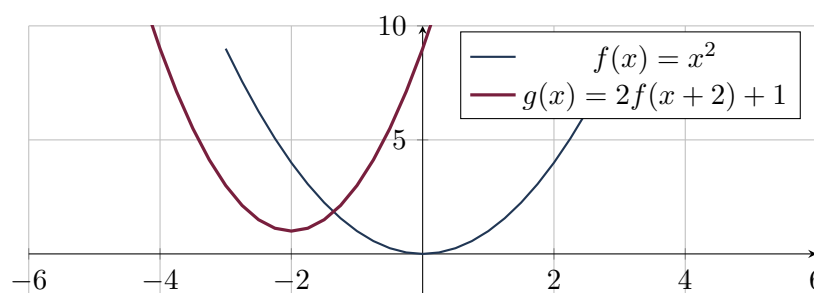
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Learning Targets

- Describe translations, reflections, stretches, and compressions.
- Use preimage-image language to track points through transformations.
- Distinguish horizontal transformations from vertical transformations.

Essential Question

How does a symbolic change to a function rule move every point on the graph?



Concept Framework

Key Idea

For $g(x) = af(b(x - h)) + k$, input transformations occur inside and output transformations occur outside.

Key Idea

Horizontal scale is reciprocal: b produces factor $1/|b|$.

Key Idea

A negative a reflects across the x-axis; a negative b reflects across the y-axis.

Key Idea

Track a point $(u, f(u))$ by solving $b(x - h) = u$.

Formula Map**Formula and Structure**

$$g(x) = af(b(x - h)) + k$$

Formula and Structure

$$(u, f(u)) \mapsto \left(h + \frac{u}{b}, af(u) + k\right)$$

Worked Examples**Worked Example 1**

Problem. Describe $g(x) = -3f(2(x - 4)) + 5$.

Solution. Horizontal compression by factor $1/2$, shift right 4, reflect across the x-axis, vertical stretch by 3, shift up 5.

Worked Example 2

Problem. If $(6, -2)$ lies on f , find the corresponding point on $g(x) = 2f(x + 1) - 3$.

Solution. Solve $x + 1 = 6$, so $x = 5$. Output is $2(-2) - 3 = -7$. Point: $(5, -7)$.

Worked Example 3

Problem. Why does replacing x by $x - 5$ shift right rather than left?

Solution. To feed the old input u into f , the new x must satisfy $x - 5 = u$, so $x = u + 5$.

Multiple-Representation Connection**AP Reasoning**

For this topic, always connect at least two representations: algebraic structure, numerical evidence, graphical behavior, or contextual meaning. State restrictions and units before interpreting a result. A correct calculation without a valid domain or contextual interpretation is incomplete AP reasoning.

Common Errors to Avoid

Common Error Check

- Do not discard domain restrictions when rewriting an expression.
- Do not infer local behavior from only one interval-wide average.
- Do not report a numerical answer without units or contextual meaning when quantities are named.
- Do not use a graphing window as proof that a feature does not exist.

Guided Check

1. Describe the transformations in $g(x) = f(x - 4) + 3$.

2. Describe the transformations in $g(x) = -2f(x + 1)$.

3. Describe the transformations in $g(x) = f(3x)$.

4. Describe the transformations in $g(x) = f(-x) + 5$.

5. Describe the transformations in $g(x) = \frac{1}{2}f(x/4) - 2$.

6. Point $(-2, 5)$ lies on f . Find its image under $g(x) = -3f(2(x - 1)) + 4$.

Lesson Summary

Complete these sentences before leaving the lesson:

1. The most important structure in this topic is _____.
2. A restriction I must remember is _____.
3. One graphical feature I can justify algebraically is _____.